Scenario Based Interview

Pyspark vs Spark SQL

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#Problem Statement---> Remove all reversed number pairs from given table, keep only one (random) if something exists...

```
from pyspark.sql import SparkSession
from pyspark.sql import Row
from pyspark.sql import functions as F
# Initialize SparkSession
spark = SparkSession.builder.appName("NumberPairs").getOrCreate()

# Create data as list of Row objects
data = [
    Row(A=1, B=2),
    Row(A=3, B=2),
    Row(A=2, B=4),
    Row(A=2, B=1),
    Row(A=5, B=6),
    Row(A=4, B=2)
]

# Create DataFrame
df = spark.createDataFrame(data)

# Show DataFrame
df.show()

+---+---+
|  A|  B|
+---+---+
|  1|  2|
|  3|  2|
|  2|  4|
|  2|  1|
|  5|  6|
|  4|  2|
+---+---+


df = df.withColumn("OrderedPair", F.array_sort(F.array("A", "B")))

# Select distinct pairs
distinct_pairs_df = df.select(
    F.col("OrderedPair")[0].alias("A"),
    F.col("OrderedPair")[1].alias("B")
).distinct()

# Show the resulting DataFrame
distinct_pairs_df.display()

# Register DataFrame as a SQL temporary view
df.createOrReplaceTempView("number_pairs")
```

```python
# Use Spark SQL to query the temporary view
spark.sql("SELECT * FROM number_pairs").show()
```

```
+---+---+
|  A|  B|
+---+---+
|  1|  2|
|  3|  2|
|  2|  4|
|  2|  1|
|  5|  6|
|  4|  2|
+---+---+
```

```python
# Save DataFrame as a table in Spark's managed table format
df.write.saveAsTable("number_pairs")

# Query the table using Spark SQL
spark.sql("SELECT * FROM number_pairs").show()
```

```
+---+---+
|  A|  B|
+---+---+
|  1|  2|
|  3|  2|
|  2|  4|
|  2|  1|
|  5|  6|
|  4|  2|
+---+---+
```

```sql
%sql
with cte1 as(
        select a, b
        , case
            when a>b then a||b
            else b||a
        end as grouping
        from number_pairs)
        ,cte2 as(
        select a,b, grouping
        , row_number() over(partition by grouping order by a) row_num
        from cte1)
select a,b
from cte2
where row_num = 1
```

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